National Aeronautics and Space Administration

RealWindDroneSim (RWDS)

Enhancing Simulation Fidelity of small Uncrewed Aerial Vehicles in Windy Conditions

Challenge

- State-of-the-art sUAS simulation platforms such as Gazebo and Airsim often use over simplified wind models, and do not have environment awareness, resulting inaccurate result.
- sUAS simulation platforms also fall short in wind simulation accuracy, and do not offer different types of wind such as turbulent, shear, or cross-winds.

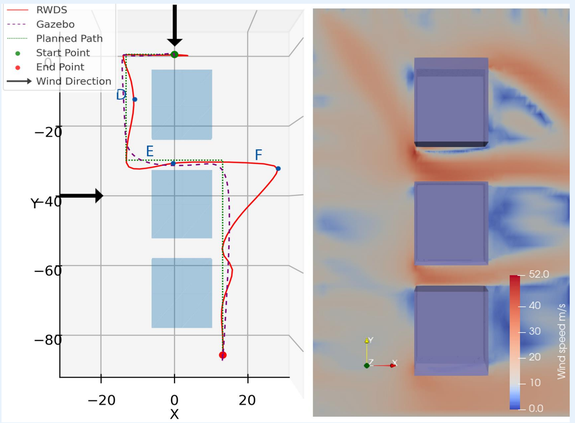

Fig 1: RWDS Wind Simulation (Right) and its impact on sUAS trajectory (Left)

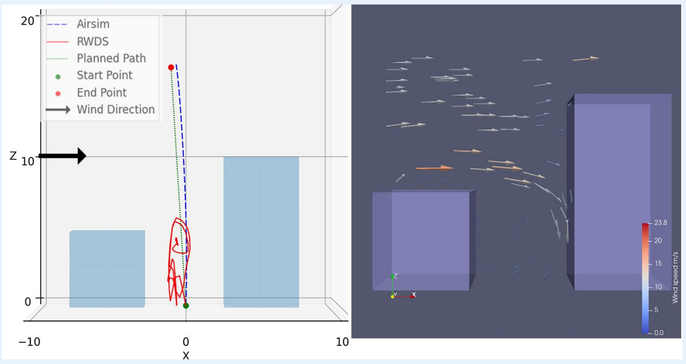

Fig 3: RWDS Identifying Failures in a Real Use-Case

Expected Impacts

- RWDS offers sUAS developers an easy-to-use platform to simulate realistic wind types and conditions in various environments.
- Improved wind simulation accuracy helps developers identify and mitigate wind related safety concerns.
- RWDS offers the sUAS community a great tool for testing purpose during the early phases of development

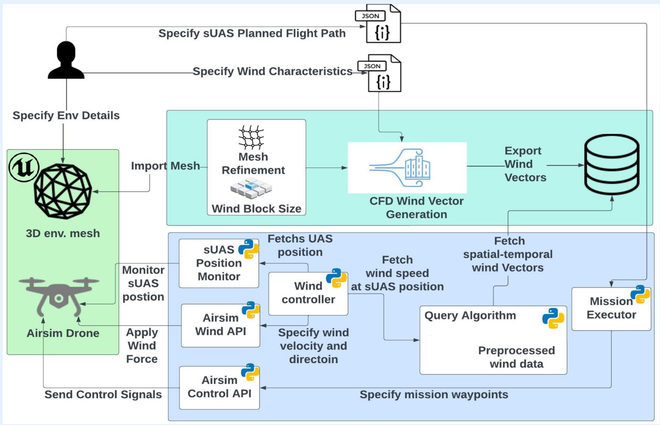

Fig 2: RWDS System Architecture

Fig 4: RWDS integrated with Unreal Engine for High Fidelity Simulations

Proposed Solution

- **Computational Fluid Dynamics (CFD) based simulation:** We applied Computational CFD principles, utilizing the Navier-Stokes equations, to generate precise wind velocity vectors (Direction, Velocity, Space) in any given 3D environment.
- **System Architecture:** We proposed a novel system architecture (**Fig 2**) to preprocess the 3D environment mesh and integrate the CFD based wind velocity vectors in sUAS simulations.
- **High Fidelity Simulation**: We integrated RWDS with Unreal Game Engine to simulate realistic wind in city scale digital twin models (**Fig 4**).

Results

- **Realistic Impact of Wind on sUAS Flight Path: Fig1** illustrates that the RWDS produced wind vectors according to the environment's geometry (Right), with the corresponding effect observed on the sUAS flight path (Left). In contrast, Gazebo simulated the sUAS flight path while ignoring the impact of blocks on wind flows, as well as the resulting sUAS flight path.
- **Identifying Wind-Related Failures: Fig. 3** shows that RWDS successfully simulated a whirlwind between two buildings and a crash landing, whereas Airsim lacked the ability to simulate such dynamic behavior of wind.

Next Steps

- Currently, RWDS has only been compared with existing sUAS simulation platforms, Gazebo and Airsim. We are now working to collect sUAS flight data under specific wind conditions in the real world and compare these results with the simulated outcomes.

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IMAGINAVIATION